



Pacific  
Community  
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# Scoping the Next Generation of Tuna Stock Assessment Software (P123)

*External Tagging Analysis and Assessment Model Design*

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SPC PAW Meeting

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# Overview

**External Tagging Analysis**     *method being developed by DTU scientists,  
related but different from SEAPODYM*

**Assessment Model Design**     *SPC – IATTC – NOAA – Quantifish collaboration,  
RTMB estimation platform, time-varying selectivity*

**SC22 Project Proposal**     *budget, work plan, timeline*

## External tagging analysis

Method being developed by [DTU scientists](#), in collaboration with SPC and IATTC

The current analysis focuses on WCPO skipjack tags, with the intention to analyze yellowfin and bigeye tags later

Initial workshop in 2025

- Tobias Mildenberger, Anders Nielsen (DTU)
- Inna Senina, Joe Scutt Phillips, John Hampton, Arni Magnusson (SPC)
- Mark Maunder, Rujia Bi (IATTC)

Currently securing funding for this work to proceed

- small amount for 2026 (from P123 budget)
- larger amount for 2027 (from upcoming Software Development Project budget)
- method development could be completed in 2028 *for that year's SKJ assessment*

## External tagging analysis

Related but different from SEAPODYM

### DTU Movement Model

- Uses all tags, also those that are not recaptured

- Designed to produce abundance indices

- Software maintained by Tobias Mildenerger (DTU) using RTMB

### SEAPODYM

- Only uses recaptured tags

- Designed to estimate fish movement

- Software maintained by Inna Senina (SPC) using AUTODIF C++ libraries

# Assessment model design

## CAPAM 2025

Mini-conference showcasing cutting-edge research across many research domains in stock assessment modelling

<https://www.populationassessment.com/tuna/next-gen-model/workshop>

## Priority features

- ▶ Features that are currently used in WCPFC tuna stock assessments
- + Time-varying selectivity, using random effects
- No tagging analysis inside the model

# Development paradigm and first steps

## Continuous and dynamic

- Design decisions will be made based on findings during development
- Mark Maunder contributes to model design coordination

## OPAL project

- Nicholas Ducharme-Barth (NOAA), Darcy Webber (Quantifish), Philipp Neubauer (Dragonfly)

<https://github.com/N-DucharmeBarth-NOAA/opal>

# SC22 project proposal

## Moving from recommendations to software development

The plan is to deliver a [WCPFC project proposal](#) to SC22 in 2026, to follow up on recommendations from the scoping project

This will include a proposed budget, work plan, and timeline

# Work plan

The scoping project is currently working with domain experts to draft a work plan for the project that will be proposed at SC22

One work plan for developing the [software for tagging analysis](#) and another work plan for developing the [tuna assessment model software](#)



# Timeline

## DTU Movement Model

| Year | Activity                                                                           | Amount required |
|------|------------------------------------------------------------------------------------|-----------------|
| 2026 | Demonstrate method at SC22<br>Contribute to SC22 Project Proposal                  | Small           |
| 2027 | Enhance statistical method and software<br>Deliver abundance indices for SKJ assmt | Medium          |
| 2028 | Finalize SKJ abundance indicies<br>Help SPC analyze YFT and BET indices            | Medium          |
| 2029 | Finalize YFT and BET indices                                                       | Small           |

# Timeline

## Tuna Assessment Model

| Year | Activity                                                                                                  | Amount required |
|------|-----------------------------------------------------------------------------------------------------------|-----------------|
| 2026 | Develop prototype model components<br>Demonstrate software at SC22<br>Contribute to SC22 Project Proposal | Medium          |
| 2027 | Develop basic tuna assessment model<br>Test model on single-region BET dataset                            | Large           |
| 2028 | Develop prioritized model features<br>Test model on SKJ dataset                                           | Large           |
| 2029 | Develop prioritized model features<br>Test model on YFT and BET datasets                                  | Large           |

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These slides:

<https://github.com/PacificCommunity/ofp-sam-transition-plan>